

CLIENT REQUIREMENTS

CMPG325-2026-104 | Bongani Nkosi | 37786458

1. ORGANISATION OVERVIEW

Client: Ramotshere Moiloa Local Municipality Offices (Zeerust)
Industry: Municipal Services
Location: Zeerust, North West Province

2. PROJECT BACKGROUND

The Ramotshere Moiloa Local Municipality requires a computer network to support its administrative functions. The network must provide connectivity across departments while maintaining security and segmentation.

3. ASSUMPTIONS MADE

Assumption	Justification
5 Departments exist	Typical municipal structure
~75 Users total	Based on town population (~15,000)
CCTV system in place	Security requirement for municipal buildings

4. REQUIREMENTS SUMMARY

Requirement	Detail
Connectivity	All departments must communicate internally
Security	CCTV segmented from office traffic
Internet Access	NAT required for external access
Server Access	Only authorised departments can access servers

5. DEPARTMENTAL STRUCTURE

Department	Users	Functions
Administration	15	Management, legal, registry
Finance	12	Budgeting, payroll, procurement
Human Resources	8	Staff records, recruitment
Public Services	20	Customer care, licensing
IT/Technical	6	Network support
Security/CCTV	4	Camera monitoring

6. JUSTIFICATION OF ASSUMPTIONS

All assumptions are based on:

- Standard South African municipal structures
- Typical departmental functions
- Reasonable scale for a town of Zeerust's size
- Common network design best practices

PHYSICAL TOPOLOGY

CMPG325-2026-104 | Bongani Nkosi | 37786458

1. PHYSICAL LAYOUT

Building 1: Main Administration Building

- **Departments:** Admin, Finance, HR, IT
- **Devices:** 3x Access Switches (Cisco 2960)
- **Users:** ~41 staff

Building 2: Public Services Centre

- **Department:** Public Services
- **Devices:** 1x Access Switch (Cisco 2960)
- **Users:** ~20 staff

Building 3: IT/Server Room

- **Devices:** 1x Core Switch (3650), 1x Edge Router (4321)
- **Servers:** 2x Servers (App + File)

Building 4: CCTV Building

- **Devices:** 1x Access Switch (2960), 8-12 Cameras
- **Purpose:** Security monitoring

2. DEVICE INVENTORY

Device	Model	Quantity	Location	Purpose
Edge Router	Cisco 4321	1	Server Room	NAT/Gateway

Device	Model	Quantity	Location	Purpose
Core Switch	Cisco 3650	1	Server Room	L3 Routing
Access Switch	Cisco 2960	4	Various Buildings	End user access
Server	Generic Server	2	Server Room	App/File hosting
PC	Generic PC	75	Various	End users
CCTV Camera	IP Camera	8-12	CCTV Building	Security

3. CABLING PLAN

Connection	Medium	Speed
Core to Access	Fiber	1 Gbps
Access to End Devices	Copper (Cat6)	100 Mbps
Router to ISP	Copper	100 Mbps

LOGICAL TOPOLOGY

CMPG325-2026-104 | Bongani Nkosi | 37786458

1. VLAN DESIGN

VLAN ID	VLAN Name	Subnet	Gateway	Purpose
10	ADMIN	10.40.1.0/24	10.40.1.254	Administration
20	FINANCE	10.40.2.0/24	10.40.2.254	Finance Dept
30	HR	10.40.3.0/24	10.40.3.254	HR Dept

VLAN ID	VLAN Name	Subnet	Gateway	Purpose
40	PUBLIC	10.40.4.0/24	10.40.4.254	Public Services
50	IT	10.40.5.0/24	10.40.5.254	IT Dept
100	CCTV	10.40.10.0/24	10.40.10.254	CCTV (Segmented)
200	SERVERS	10.40.20.0/28	10.40.20.254	Server Farm
999	MANAGEMENT	10.40.255.0/28	10.40.255.254	Network Mgmt

2. ROUTING DESIGN

Device	Routing Type	Routes
Edge Router	Static Default	0.0.0.0/0 → 192.168.1.1
Edge Router	Static	10.40.0.0/16 → 10.40.1.254
Core Switch	Static Default	0.0.0.0/0 → 10.40.1.1

3. SECURITY ACLs

ACL 101: CCTV Segmentation

```
access-list 101 deny ip 10.40.10.0 0.0.0.255 10.40.0.0 0.0.255.255
access-list 101 permit ip any any
```

Template 4: IP-Addressing-Plan.md

```
```markdown
IP ADDRESSING PLAN {#ip-addressing-plan }
CMPG325-2026-104 | Bongani Nkosi | 37786458 {#cmpg325-2026-104--bongani-nkosi--37786458-3 }
```

---

## 1. ADDRESSING OVERVIEW {#1-addressing-overview }

```
Assigned Block: 10.40.0.0/16
Total Addresses: 65,536
```

---

## 2. SUBNET ALLOCATION {#2-subnet-allocation }

Subnet	Range	Gateway	Hosts	VLAN
10.40.1.0/24	10.40.1.1-254	10.40.1.254	254	10
10.40.2.0/24	10.40.2.1-254	10.40.2.254	254	20
10.40.3.0/24	10.40.3.1-254	10.40.3.254	254	30
10.40.4.0/24	10.40.4.1-254	10.40.4.254	254	40
10.40.5.0/24	10.40.5.1-254	10.40.5.254	254	50
10.40.10.0/24	10.40.10.1-254	10.40.10.254	254	100
10.40.20.0/28	10.40.20.1-14	10.40.20.254	14	200
10.40.255.0/28	10.40.255.1-14	10.40.255.254	14	999

---

## 3. DEVICE IP ASSIGNMENTS {#3-device-ip-assignments }

Device	Interface	IP	Mask
Edge Router	G0/0 (Inside)	10.40.1.1	/24
Edge Router	G0/1 (Outside)	192.168.1.2	/30
Core Switch	VLAN 10	10.40.1.254	/24
Core Switch	VLAN 20	10.40.2.254	/24
Core Switch	VLAN 30	10.40.3.254	/24
Core Switch	VLAN 40	10.40.4.254	/24
Core Switch	VLAN 50	10.40.5.254	/24
Core Switch	VLAN 100	10.40.10.254	/24
Core Switch	VLAN 200	10.40.20.254	/28
App Server	NIC	10.40.20.2	/28
File Server	NIC	10.40.20.3	/28

---

## 4. DHCP RANGES {#4-dhcp-ranges }

VLAN	DHCP Range	Gateway	DNS
10	10.40.1.10-10.40.1.240	10.40.1.254	8.8.8.8
20	10.40.2.10-10.40.2.240	10.40.2.254	8.8.8.8
30	10.40.3.10-10.40.3.240	10.40.3.254	8.8.8.8
40	10.40.4.10-10.40.4.240	10.40.4.254	8.8.8.8
50	10.40.5.10-10.40.5.240	10.40.5.254	8.8.8.8

---

## 5. NAT MAPPINGS {#5-nat-mappings }

NAT Type	Inside	Outside	Purpose
PAT	All Office Subnets	192.168.1.2	Internet Access
Static	10.40.20.2	192.168.1.3	App Server
Static	10.40.20.3	192.168.1.4	File Server